

Faculty of Electrical Engineering

WARSAW UNIVERSITY OF TECHNOLOGY

1DA1636:A Automotive Data Transmission Systems

PROJECT

TITLE

1-WIRE TEMPERATURE SENSOR WITH ALERT FUNCTION

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1. Introduction

The objective of the project is to create a working system from 1-wire temperature sensor which will let us observe the temperature in real time and alarm(notify) us when temperature goes certain thresholds. As a part of the project libraries with reference are allowed. OLED display is used to display the temperature along with serial monitor. And finally passive buzzer which can be set to the temperature of interest to be alerted.

My choice was influenced by my background. I took a course in conditioning sensor signals. But found out very fast that theoretical knowledge isnt much needed thing for this project. Rather hands on experience would have helped the most. That is the thing I lacked, this is my first project in hands in electronics.

2. Literature review

The starting point of the project is searching for 1-wire sensor. After finding the most used/common one is DS18B20, next step is to loop up the documentation of it [1]. This is the place the information on type(analog/digital, it is digital) of pin to be connected and the circuit schematic was taken. Also, since the libraries are allowed their readme, examples parts on github can probably considered as a literature, too. So here OneWire library[2] and DallasTemperature[3] are must-have libraries if we want to write the least code. [3] can be considered the library we need and [2] is what it can not work without, meaning [3] which is built for specifically the temperature sensors with 1-wire works on OneWire library.
It can be stated that [6] is the perfect library for the OLED display used. And from its github we can see that it utilizes(works on) Wire[4] and AdafruitGFX[5] libraries.

[1] Documentation DS18B20
[2] OneWire library
[3] DallasTemperature library and example
[4] Wire library
[5] AdafruitGFX library
[6] AdafruitSSD1306 library and example

3. Methodology

Hardware and software used:.

Microcontroller used - Arduino Uno R4.
Temperature sensor - DS18B20 (one wire temperature sensor).
Jumper wires, resistor, OLED display with SSD1306 processor type, passive buzzer and bread boards.
Software is Arduino IDE.

Methodology was to buy initially suitable components, since this was new for me it took some time. And then searching the circuit to connect the 1-wire system. Which is found from documentation Step by step process is very important concept here. First temperature sensor DS18B20 was connected, results were observed. Only after that I did the screen connection and after that was successful buzzer was implemented. Each time the libraries of interest was researched to learn how to use them.

The most complex part is the circuit for the sensor itself, as illustrated in figure 1, though in general it is considered to be easy. Connecting the OLED screen is much easier. The protocol is obvious from the pin names(named SDA, SLK). The buzzer is even simpler just connecting the positive to power and negative to ground is enough

When it comes to the coding part libraries really helped. If using them wasnt allowed, i think it would have been a project of a higher level. Really all the smart things, reading from the 0s and 1s from the sensor and etc for displaying are all being done by the libraries. So the task was learning how to utilize them properly as needed.

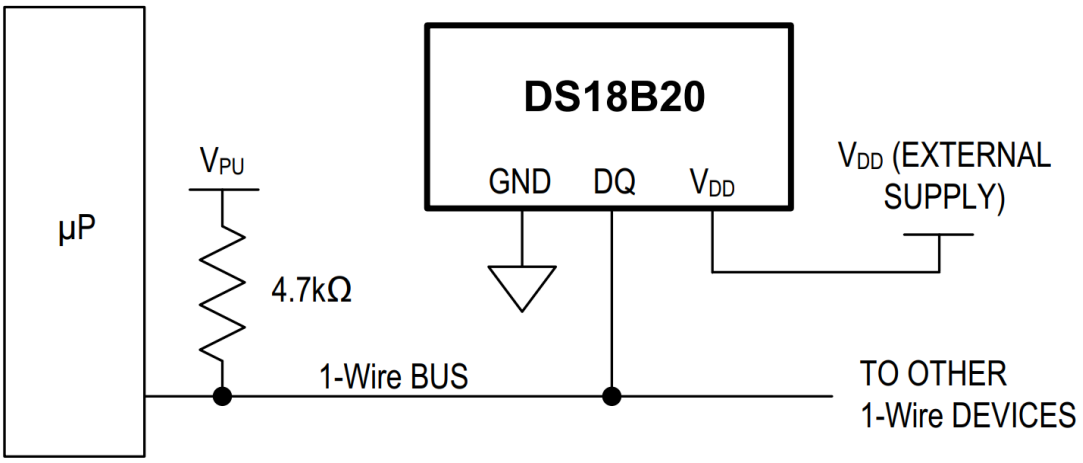


Figure 1 - connection for DS18B20 sensor(normal operation)

4. Results and Discussion

Show and discuss your results (figures, photographs, diagrams, waveforms, examples of calculations etc.). Photos, screenshots
There is two of ways viewing the result. One is on OLED and other is on serial monitor. Figure 2 shows the circuit schematic entirely except the arduino being used to supply positive line with 5 V and negative line with ground. Also digital pin 8 is connected to positive of buzzer. In figure 3 where I did a small test to see if the buzzer is working as needed. The temperature from which the alarm is going

on was set to 31, since the room temperature was around 30 degrees. The result - everything is working good.

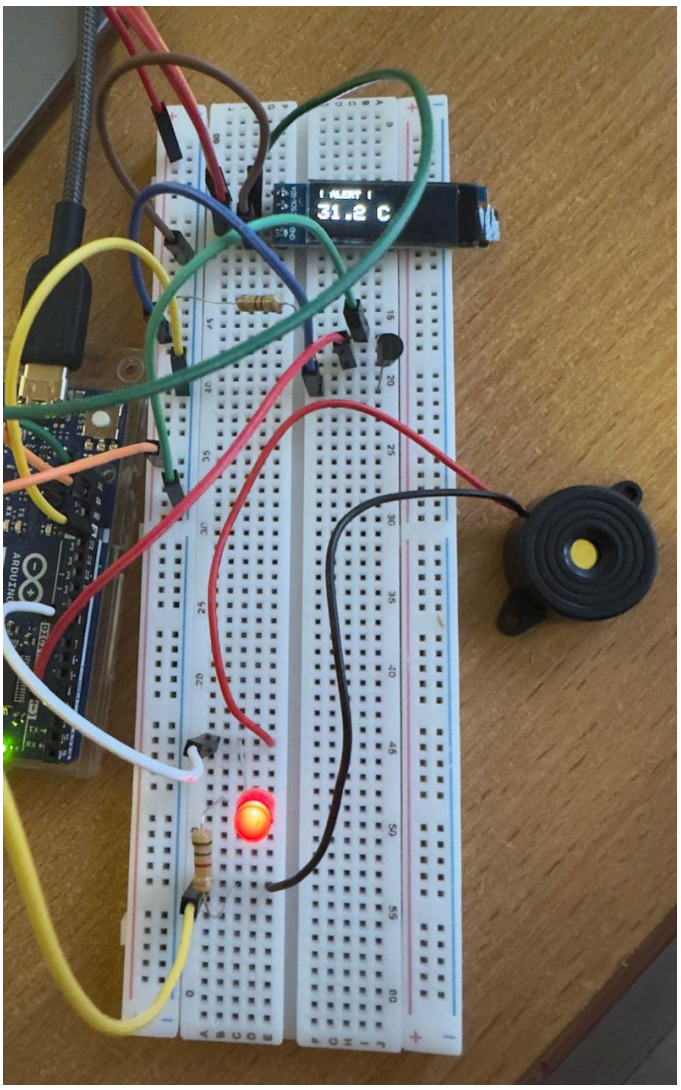


Figure 2 - The final circuit done using breadboard

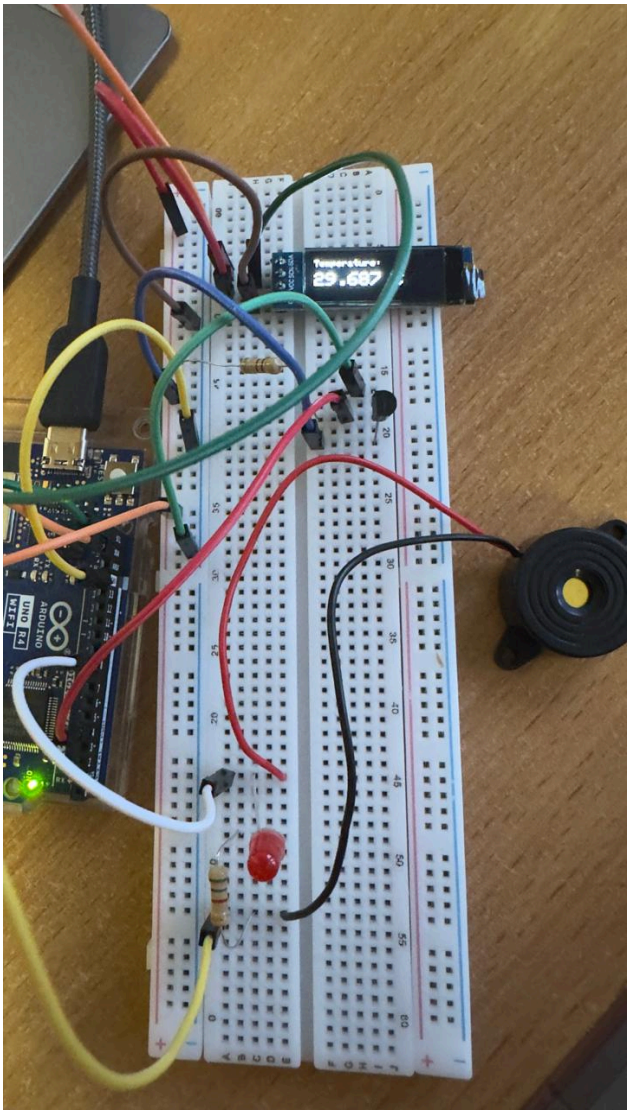


Figure 3 - when alert function is Off

```
Output  Serial Monitor ×
Message (Enter to send message to 'Arduino UNO R4 WiFi' on 'COM7')   New Line   115200 baud
19:16:35.418 -> Temperature: 30.187 C
19:16:38.032 -> Temperature: 30.687 C
19:16:40.718 -> Temperature: 31.812 C
19:16:43.369 -> Temperature: 32.500 C
19:16:45.994 -> Temperature: 32.938 C
19:16:48.679 -> Temperature: 33.188 C
19:16:51.336 -> Temperature: 32.750 C
19:16:53.974 -> Temperature: 32.313 C
19:16:56.654 -> Temperature: 32.000 C
19:16:59.264 -> Temperature: 31.687 C
19:17:01.919 -> Temperature: 31.500 C
19:17:04.617 -> Temperature: 31.375 C
19:17:07.226 -> Temperature: 31.250 C
19:17:09.904 -> Temperature: 31.125 C
19:17:12.572 -> Temperature: 31.062 C
19:17:15.188 -> Temperature: 31.000 C
19:17:17.877 -> Temperature: 30.937 C
```

Figure 4 - serial monitor screenshot. Sensor was heated by touching by hand and then cooled after reaching 32.750 C.

Lastly, here is the buzzer function which turns buzzer on when current temperature is higher than what we have set.

```
if (tempC > temp_alert) {
```

```
    tone(buzzer, 100);
```

```
  } else {

    noTone(buzzer);

  }
```

5. Conclusions

Are all goals achieved? Were all of them achievable? What was difficult, not completed etc. Why? Future work? What was successful, troubles, problems

It is safe to say that all goals were achieved.
The problematic thing which hindered my progress for weeks was the Not soldered pins of my microcontrollers. I had 2 microcontroller tried changing them. Not only them I tried changing everything in the circuit - resistors, breadboards, wires, sensors(I have 4 sensors now). So the issue was solved by buying Arduino.

References

[1] Documentation DS18B20 - https://www.analog.com/media/en/technical-documentation/data-sheets/ds18b20.pdf
[2] OneWire library - https://github.com/PaulStoffregen/OneWire/blob/master/examples/DS18x20_Temperature/DS18x20_Temperature.ino
[3] DallasTemperature library and example - https://github.com/milesburton/Arduino-Temperature-Control-Library
[4] Wire library
[5] AdafruitGFX library
[6] AdafruitSSD1306 library and example - https://github.com/adafruit/Adafruit_SSD1306/blob/master/examples/ssd1306_128x64_i2c/ssd1306_128x64_i2c.ino